

Lab 5 - Experimental Design

ENVX2001 Applied Statistical Methods

Semester 1, 2026

Learning outcomes

At the end of this lab students should be able to:

- distinguish between sampling units and experimental units;
- use R to generate randomisations for CRDs and RCBDs;
- use R to analyse experiments with blocking, and assess the usefulness of blocking.

All of the data for this lab is in the **Data5.xlsx** file.

Exercise 1 - Randomisation for a CRD using R (Walk-through)

Consider a glass house experiment conducted on a bench on which it was judged that the growing conditions would be consistent across the bench. The experimenter had five different smoked water treatments for assisting the germination of *Banksia* seeds. She had 50 dishes on which she could place seeds randomly selected from a uniform batch of *Banksia* seeds. The dishes were to be placed on the bench and each of the five water solutions allocated randomly to 10 of the dishes.

Question 1.1

(i) What would the degrees of freedom be in a corresponding ANOVA table?

Question 1.2

(ii) Suppose that Treatment 1 was a control. In situations where the comparisons of interest are of the other treatments (2 to 5) with the control, and not amongst the other treatments, it can be advantageous to increase the number of control replicates to 20. What are the degrees of freedom in the corresponding ANOVA table?

Question 1.3

(iii) The formula for standard error of the difference (SED) is:

$$SED = \sqrt{Resid\ MS\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}.$$

Given the SED formula, what percentage improvement (i.e. reduction) has been achieved for the SED between Treatment 1 and 2 when there are equal replicates (i) versus unequal replicates (ii)? Assume the Residual MS is a constant.

Exercise 2 - The balance between the number of sampling units and experimental units (Walk-through)

Horticulturalists have been studying the nutrient requirements of lettuce growing in a sand medium; in particular they are looking at the nitrate concentration in leaf petioles in response to varying applied nitrogen nutrient levels (5, 11, 18 and 32 mmol/L). Lettuce plants are grown in separate pots, so different nitrogen levels may be applied to individual pots, and one or more leaves sampled per plant. They are also interested in optimising their experimental protocol for future studies; consequently they conducted five separate experiments where they sampled different number of leaves per plant, and differing number of plants, but kept the total number of leaves sampled at 128 in each experiment. The experiments were conducted as follows:

- Experiment 1: 16 leaves per plant; 2 plants per treatment;
- Experiment 2: 8 leaves per plant; 4 plants per treatment;
- Experiment 3: 4 leaves per plant; 8 plants per treatment;
- Experiment 4: 2 leaves per plant; 16 plants per treatment;
- Experiment 5: 1 leaf per plant; 32 plants per treatment.

Similar lettuce plants were used across all five experiments, and in all cases, a CRD was used to allocate the plants amongst the treatments. The datasets are located in the *Data5.xlsx* file. There are separate sheets for each Experiment called *Experiment1*, *Experiment2*, etc.

Question 2.1

(i) What are the experimental units and what are the sampling units in the above experiments. For each experiment, how many experimental units and how many sampling units are used?

Question 2.2

(ii) When there are multiple sampling units per experimental unit, in simple situations (as is here), an appropriate method is simply to average the responses over all the sampling units belonging to an experimental unit, and analyse these. So in this case, we need to average the observed nitrate values over all the leaves in the plant. The code below shows how to do this

using a function called `rowMeans` which finds the average in a row across multiple columns. Here we do this for *Experiment1*.

```
CODE
#
```

Question 2.3

(iii) Perform a one-way ANOVA on these average data. (It will also be helpful to obtain the treatment means using the `emmeans` function, as done in previous topics. What are your conclusions about the effect of varying applied nitrogen levels? You don't have to look at post-hoc tests, just examine the F-test.

```
CODE
#
```

Question 2.4

(iv) Repeat this for Experiments 2 to 5. What are your conclusions in each experiment?

Experiment 2: 8 leaves per plant, 4 plants per treatment

```
CODE
#
```

Experiment 3: 4 leaves per plant, 8 plants per treatment

```
CODE
#
```

Experiment 4: 2 leaves per plant, 16 plants per treatment

```
CODE
#
```

Experiment 5: 1 leaf per plant, 32 plants per treatment. Note, no averaging required as sampling unit = experimental unit.

```
CODE
#
```

Question 2.5

(v) Summarise the results in terms of the balance between sampling plants versus leaves, and what recommendations would you make for future studies?

```
CODE
```

#

Question 2.6

(vi) There are two sources of random variation encountered in these experiments:

- variation between leaves within a plant; and
- variation between plants within the same treatment group.

How do you think your recommendations might change in (iv) if:

- there was (virtually) no variation between leaves within a plant; only between plants; or
- there was (virtually) no variation between plants, only variation between leaves within a plant.

Exercise 3 - A paired t-test generalises to a one-way ANOVA – RCBD

Fifteen farms cooperated in a field trial in which a normal fattening ration for pigs (ration A) was compared with the same rations supplemented with a small trace of copper (ration B). Each farmer set up two pens of pigs, as similar as possible in all respects, and allocated the two rations at random, ration A to one pen and ration B to the other. The mean weight gains per pen (g/day) are stored in the two columns of the **copper** sheet in the **Data5.xlsx** file.

This description clearly suggests the design is an RCBD, with 15 farms but only 2 treatments. As we have seen, a paired design is the simplest form of RCBD. In this example we will not worry about model assumptions.

Question 3.1

(i) Firstly analyse the data using a paired t-test. Assuming you read the data as a data frame called **copper** the code is `t.test(x=copper$RationA, y=copper$RationB, paired=T)`. Interpret the results.

CODE
#

Question 3.2

(ii) Next we will obtain an analysis of the data using ANOVA. To do this analysis however, you will first need to re-arrange the data set into a form suitable for ANOVA. That is, you will need

a single column of weight gains, a column of treatment identifiers, and a column indicating the Farm number. The code below will do this for you:

- `dat←stack(copper[,2:3])`
- `farms←rep(c(1:15),times=2)`
- `coppern←cbind(farms,dat)`

The first line stacks (using the `stack` function) the 2 columns of weight gain on top of each other (column name is `values`) and adds a column indicating which ration is associated with each weight gain value (column name is `ind`).

The second line repeats (using the `rep` function) the values of farm numbers twice so for each of the weight gains we have an associated farm.

The third line joins the two data frames together (using the `cbind` function) to create the new stacked dataset which can be used by the `aov` function.

CODE

```
head(copper)
```

OUTPUT

```
# A tibble: 6 × 3
  Farm RationA RationB
  <dbl>   <dbl>   <dbl>
1     1     422     531
2     2     526     467
3     3     476     558
4     4     499     585
5     5     422     472
6     6     503     522
```

CODE

```
dat←stack(copper[,2:3])
farms←rep(c(1:15),times=2)
coppern←cbind(farms,dat)
str(coppern)
```

OUTPUT

```
'data.frame':  30 obs. of  3 variables:
 $ farms : int  1 2 3 4 5 6 7 8 9 10 ...
 $ values: num  422 526 476 499 422 503 445 449 299 517 ...
 $ ind   : Factor w/ 2 levels "RationA","RationB": 1 1 1 1 1 1 1 1 1 1 ...
```

Question 3.3

(iii) The general code for performing an 1-way anova with a randomised complete block design is:

```
aov(response~block+treatment,data=data)
```

In this example the response is the weight gain, blocks are farms and the treatment is ration. Your names will be different. Having obtained the analysis, check that the results from the ANOVA is identical to the paired t -test:

- the same P -value;
- the same test statistic (though expressed as an F -statistic instead of a t -statistic, with $F = t^2$ to that from the paired t -test.

```
CODE
#
```

Exercise 4 - One-way ANOVA - RCBD

Three diets for hamsters were tested for differences in weight gain after a specified period of time. Six inbred lines were used with three hamsters selected from each line. The three diets were assigned at random to the hamsters in each line. The data is in the **hamsters** sheet in the **Data5.xlsx** file.

Question 4.1

(i) Are the mean weight gains the same across diets? Are any diets more or less effective than other diets? You will need to use the **emmeans** package and its **emmeans()** function. Read in the data and perform post-hoc tests.

```
CODE
#
```

Question 4.2

(ii) Re-run the analysis without blocking, i.e. as a CRD. Are the conclusions different? Would you use blocking in the future? What proportion of the variation was explained by blocking?

```
CODE
#
```